

Credit Card Fraud Detection System using Machine Learning

Abstract

This project develops a machine learning web application to detect fraudulent credit card transactions. Multiple classification algorithms were trained and evaluated before deploying the best model using Flask and Render.

Objectives

Detect fraudulent transactions accurately, reduce manual effort, and provide real-time predictions.

Dataset

The project uses the Credit Card Fraud Detection dataset containing anonymized transaction features, Time, Amount, and Class.

Technologies

Python, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, XGBoost, Flask, HTML, CSS, Joblib, GitHub, Render.

Methodology

Performed preprocessing, EDA, feature scaling, model training, evaluation, best model selection, model serialization, Flask integration, and deployment.

Models

Logistic Regression, Decision Tree, Random Forest, and XGBoost.

Deployment

The application was deployed on Render and made publicly accessible through a web URL.

Conclusion

The completed system provides a user-friendly interface for predicting whether a transaction is fraudulent or legitimate.